

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		award (1) for either of following aluminium is more reactive than iron aluminium displaces iron award (1) for correct reference to reduction e.g. aluminium reduces iron(III) oxide aluminium takes the oxygen away from the iron iron(III) oxide is reduced	2			2		
		(ii)		$\text{Fe}_2\text{O}_3 + 2\text{Al} \rightarrow 2\text{Fe} + \text{Al}_2\text{O}_3$ products (1) balancing (1) balancing mark can only be awarded if the correct products given		2		2		
		(iii)		70% (2) if incorrect award (1) for any of following $M_r(\text{Fe}_2\text{O}_3) = 160$ $56 + 56 + 16 + 16 + 16$ total relative mass of iron = 112 $56 + 56$		2		2	2	